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Excavating Neandertal and Denisovan DNA from the genomes of Melanesian individuals

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Although Neandertal sequences that persist in the genomes of modern humans have been identified in Eurasians, comparable studies in people whose ancestors hybridized with both Neandertals and Denisovans are lacking. We developed an approach to identify DNA inherited from multiple archaic hominin ancestors and applied it to whole-genome sequences from 1523 geographically diverse individuals, including 35 new Island Melanesian genomes. In aggregate, we recovered 1.34 Gb and 303 Mb of the Neandertal and Denisovan genome, respectively. We leverage these maps of archaic sequence to show that Neandertal admixture occurred multiple times in different non-African populations, characterize genomic regions that are significantly depleted of archaic sequence, and identify signatures of adaptive introgression.

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For much of human history, modern humans overlapped in time and space with other hominins (*1*). Analyses of the Neandertal (*2*, *3*) and Denisovan (*4*, *5*) genomes revealed that gene flow occurred between these archaic hominins and the ancestors of modern humans (*3*–*8*). Consequently, all non-African populations derive ~2% of their ancestry from Neandertals (*3*), whereas substantial levels of Denisovan ancestry (~2-4%) are only found in Oceanic populations (*5*), although low levels of Denisovan ancestry may be more widespread (*9*, *10*). Recently, catalogs of introgressed Neandertal sequences have been created in Eurasians (*11*, *12*). However, considerably less is known about the genomic organization and characteristics of Denisovan sequence that persists in modern individuals.

We sequenced 35 individuals from 11 locations in the Bismarck Archipelago of Northern Island Melanesia, Papua New Guinea (Fig. 1A) (*13*, *14*) to a median depth of 40x (tables S1 and S2), including a trio to facilitate haplotype reconstruction. Unless otherwise noted, analyses were performed on a subset of 27 unrelated individuals (*14*). We integrated our sequencing data with SNP genotypes from 1,937 individuals spanning 159 worldwide populations (Fig. 1A and table S3) (*14*–*16*). A global Principal Component Analysis (PCA) and ADMIXTURE analysis shows that our samples cluster most closely with other Oceanic individuals (Fig. 1B and fig. S1), and population sizes inferred by PSMC are consistent with other non-African populations (fig. S2) (*14*). Furthermore, our Melanesian samples show genetic similarities to both Neandertals and Denisovans, whereas all other non-African populations only exhibit affinity toward Neandertals (Fig. 1C and fig. S3) (*14*). Using an *f4* statistic (*14*, *15*, *17*), we find significant evidence of Denisovan ancestry (Z score > 4) in our Melanesian samples, with admixture proportions varying between 1.9% and 3.4% (Fig. 1D and table S4).

Having demonstrated that our Melanesian individuals have both Neandertal and Denisovan ancestry, we developed an approach to recover and classify archaic sequence. Briefly, we first identify putative introgressed sequence using the statistic S^* (which does not use information from an archaic reference genome) (*11*, *18*) and then refine this set by comparing significant S^* haplotypes to the Neandertal and Denisovan genomes and testing whether they match more than expected by chance. Variation in neutral divergence between archaic groups across loci and incomplete lineage sorting complicate classification of archaic haplotypes as Neandertal or Denisovan (fig. S4) (*14*). To address this issue, we developed a likelihood method that operates on the bivariate distribution of Neandertal and Denisovan match P values (*14*). This framework estimates the proportion of Neandertal, Denisovan, and null sequence in the set of S^* significant haplotypes, identifies archaic haplotypes at a desired false discovery rate (FDR), and probabilistically categorizes them as Neandertal, Denisovan, or ambiguous (i.e.,

Neandertal or Denisovan status cannot be confidently distinguished) (*14*). In addition to our Melanesian samples, we also applied our method to whole-genome sequences from 1,496 geographically diverse individuals studied as part of the 1000 Genomes Project (table S5) (*14*, *19*).

We evaluated our approach through coalescent simulations (figs. S5 and S6) and by analyzing African populations (fig. S7). Archaic match *P* values calculated from null coalescent simulations without archaic admixture and significant *S** sequences in African individuals show similar distributions (Fig. 2A), consistent with little to no Neandertal or Denisovan ancestry in most African populations. Notably, Luhya and Gambians do show evidence of having some Neanderthal ancestry (fig. S8 and table S6), most likely inherited indirectly through recent admixture with non-Africans (*20*). In Europeans, we see a strong skew of Neandertal, but not Denisovan, match *P* values toward zero (Fig. 2A). In contrast, Melanesians exhibit a marked skew of both Neandertal and Denisovan match *P* values toward zero (Fig. 2A).

In aggregate, across all 1523 non-African individuals analyzed, we recovered 1340 Mb and 304 Mb of the Neandertal and Denisovan genomes, respectively, at a FDR = 5%. Melanesian individuals have on average 104 Mb of archaic sequence per individual (48.9 Mb, 42.9 Mb, and 12.2 Mb of Neandertal, Denisovan, and ambiguous sequence, respectively) (Fig. 2B). In contrast, we only call between 0.026 Mb (in Esan) to 0.5 Mb (in Luhya) of sequence per individual as archaic in Africans, highlighting that our method and error rates are well calibrated. The higher levels of archaic ancestry in Melanesians results in an average of 20 compound homozygous archaic loci per individual, with one Neandertal and one Denisovan haplotype (Fig. 2C and fig. S9). We estimate that 61% of the variability in amount of Denisovan sequence between individuals is explained by variation in Papuan ancestry ($P = 7.8 \times 10^{-4}$) (table S7) (*14*). In other non-Africans, we identify on average 65.0 Mb, 55.2 Mb, and 51.2 Mb of archaic sequence in East Asians, South Asians, and Europeans, respectively (Fig. 2B). Virtually all of the archaic sequence in these populations is Neandertal in origin, although a small fraction (<1%) of introgressed sequences in East and South Asians are predicted to be Denisovan (fig. S8 and table S6) (*14*).

We developed a method to test whether two populations have shared or unique admixture histories based on patterns of reciprocal sharing of Neandertal sequence among individuals (fig. S10) (*14*). We validated the expected behavior of our method in simulated data (Fig. 3A), and confirm previous observations of an additional admixture event unique to East Asians (*11*, *21*, *22*) (Fig. 3B). We find evidence for an additional pulse of Neandertal admixture in Europeans, East Asians, and South Asians compared to Melanesians (Fig. 3B), which is robust to different statistical thresholds used to call Neandertal and Denisovan sequence and determining significance (figs. S11 to S13) (*14*). Conversely, we find no evidence of differences in admixture histories between Europeans and South Asians (fig. S14). Collectively, these data suggest Neandertal admixture occurred at least three distinct times in modern human history (Fig. 3C). Although most South Asian populations show shared histories of archaic admixture, we find significant evidence of differential Neandertal admixture between some European and East Asian populations (figs. S15 to S17).

The density of surviving Neandertal sequence across the genome is heterogeneous (*11*), and regions that are strongly depleted of Neandertal ancestry may represent loci where archaic sequence was deleterious in hybrid individuals and purged from the population. To quantify how unusual Neandertal depleted regions are under neutral models, we performed coalescent simulations (*14*), focusing on individuals of European and East Asian ancestry whose demographic histories are known in most detail. Depletions of Neandertal sequence that extend ≥8 Mb are significantly enriched in the observed compared to simulated data (permutation $P < 0.01$) (Fig. 4A and fig. S18). Neandertal depleted regions that span at least 8 Mb are also significantly (KS test, $P < 10^{-15}$) depleted of Neandertal sequence in South Asians and Melanesians (fig. S19 and table S8).

We find significantly more overlap in regions depleted of Neandertal and Denisovan lineages than expected by chance (permutation $P = 0.0008$) (fig. S20 and table S9) (*14*), consistent with recurrent selection against deleterious archaic sequence. Indeed, deserts of archaic sequence tend to exhibit higher levels of background selection (figs. S21 and S22). Regions depleted of archaic lineages are significantly enriched for genes expressed in specific brain regions, particularly in the developing cortex and adult striatum (permutation $P < 0.05$) (table S10). A large region depleted of archaic sequence spans 11 Mb on chromosome 7 and contains the *FOXP2* gene (Fig. 4B), which has been associated with speech and language (*23*). This region is also significantly enriched for genes associated with autism spectrum disorders (Fisher's exact text, $P = 0.008$) (*14*). Although our data shows that large regions depleted of archaic ancestry are inconsistent with neutral evolution, mechanisms other than selection, such as structural variation, could also contribute to the appearance of archaic deserts and thus additional work is necessary to fully understand the origins of such regions.

We identified putative adaptively introgressed sequence in Melanesians by identifying archaic haplotypes at unusually high frequencies as determined by coalescent simulations under a wide variety of neutral demographic models (*14*). At a frequency threshold of 0.56, corresponding to the 99th percentile of simulated data, we identified 21 inde-

pendent candidate regions for adaptive introgression (Fig. 4C and table S12). Fourteen are of Neanderthal origin, three are Denisovan, three are ambiguous, and one segregates both Neanderthal and Denisovan haplotypes. Six regions do not contain any protein-coding genes, and seven high frequency archaic haplotypes span only a single gene (table S12). High frequency archaic haplotypes overlap several metabolism related genes, such as *GCG* (a hormone that increases blood glucose levels) and *PLPP1* (a membrane protein involved in lipid metabolism). Moreover, five regions either span or are adjacent to immune related genes, including a haplotype encompassing *GBP4* and *GBP7* (Fig. 4D), which are induced by interferon as part of the innate immune response.

Substantial amounts of Neandertal and Denisovan DNA can now be robustly identified in the genomes of present-day Melanesians, allowing new insights into human evolutionary history. As genome-scale data from worldwide populations continues to accumulate, a nearly complete catalog of surviving archaic lineages may soon be within reach. Key challenges remain, including evaluating the functional and phenotypic consequences of introgressed sequence and refining estimates on the timing, location, and other characteristics of admixture events. Ultimately, maps of surviving Neandertal, Denisovan, and potentially other hominin (*1*) sequences will help interpret patterns of human genomic variation and understand how archaic admixture influenced the trajectory of human evolution.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S22
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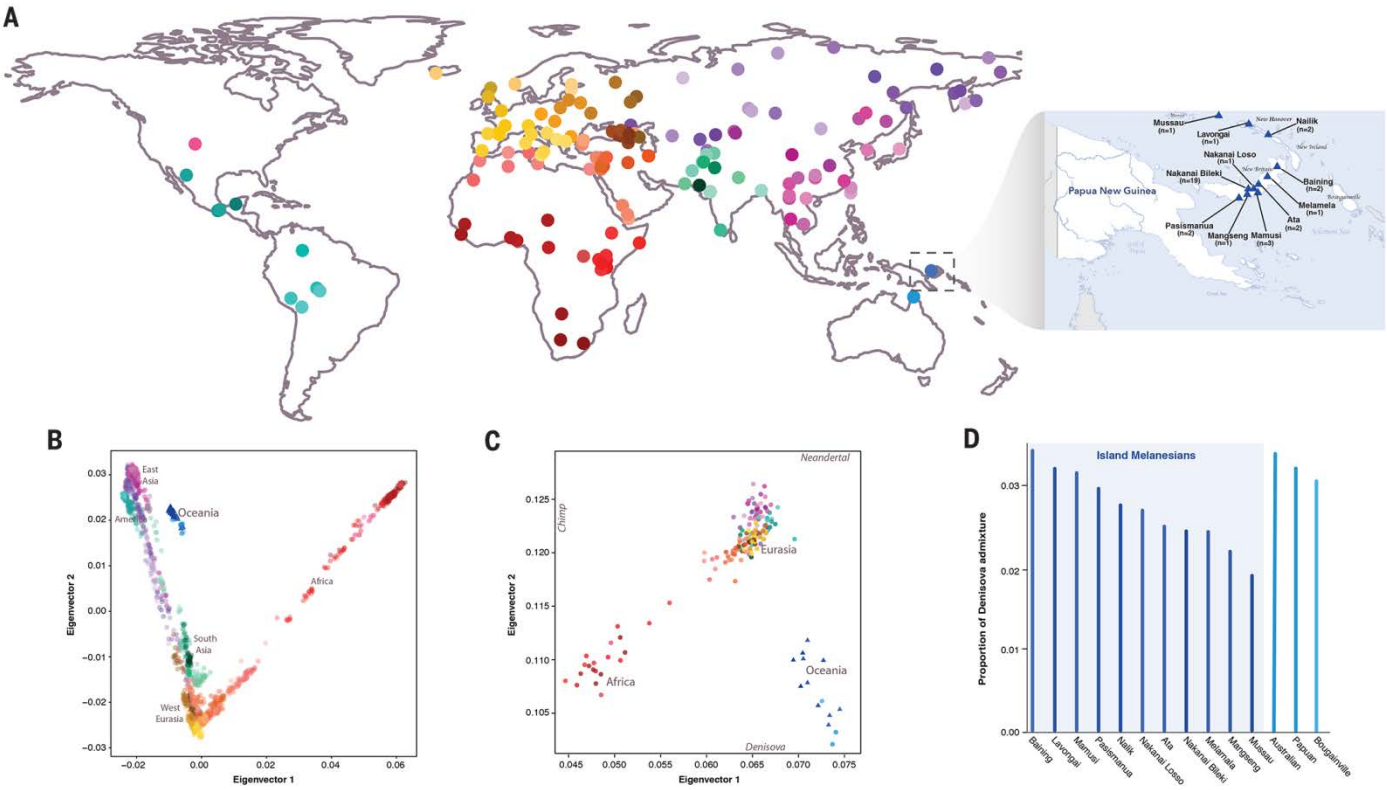


Fig. 1. Melanesian genomic variation in a global context. (**A**) Locations of the 159 geographically diverse populations studied. Information on the Melanesian individuals sequenced (blue triangles) is shown in the inset. (**B**) PCA of Melanesian genomes in the context of present-day worldwide genetic diversity. (**C**) Modern human variation projected onto the top two eigenvectors defined by PCA of the Altai Neandertal, Denisova, and chimpanzee genome (*14*). Population means were plotted for each of the 11 Melanesian populations and each population of the global dataset. (**D**) Estimates of Denisova ancestry in Oceanic populations estimated from an *f4* statistic (*14*). The 11 Melanesian populations are highlighted by the light blue box.

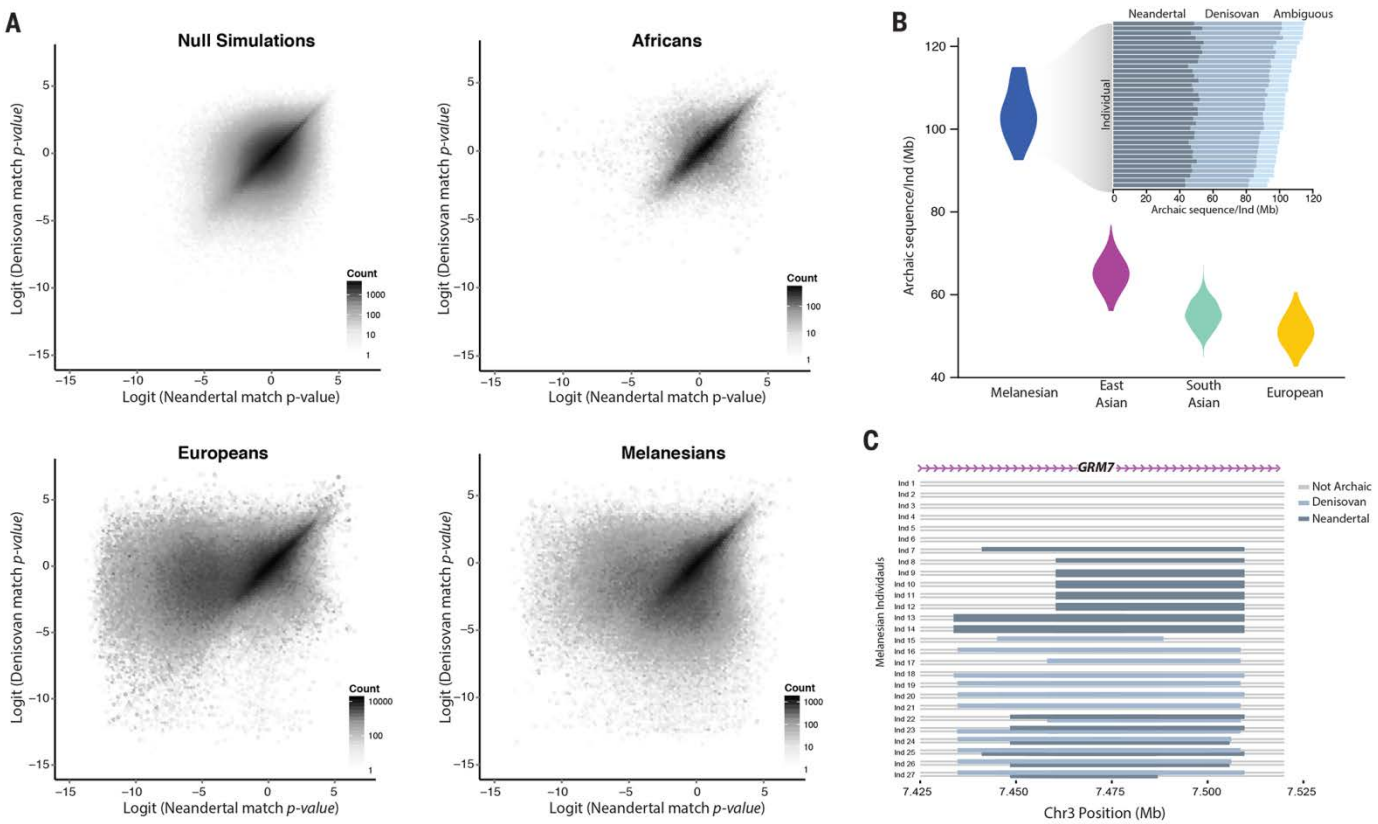


Fig. 2. Identifying Neandertal and Denisovan sequences in modern human genomes. (**A**) Bivariate archaic match *P* value distributions for simulations of non-introgressed sequence, Esan in Nigeria, Europeans, and Melanesians. Null simulations and Esan show no skew in Neandertal or Denisovan match *P* values toward zero, Europeans show only a skew of Neandertal match *P* values toward zero, and Melanesians exhibit both Neandertal and Denisovan match *P* values skewed toward zero. (**B**) Amount of archaic introgressed sequence identified in each population. Inset, amount of Neandertal, Denisovan, and ambiguous (Neandertal or Denisovan) introgressed sequence for each Melanesian individual. (**C**) Schematic representation of introgressed haplotypes in an intronic portion of the *GRM7* locus in Melanesian individuals illustrating mosaic patterns of archaic ancestry.

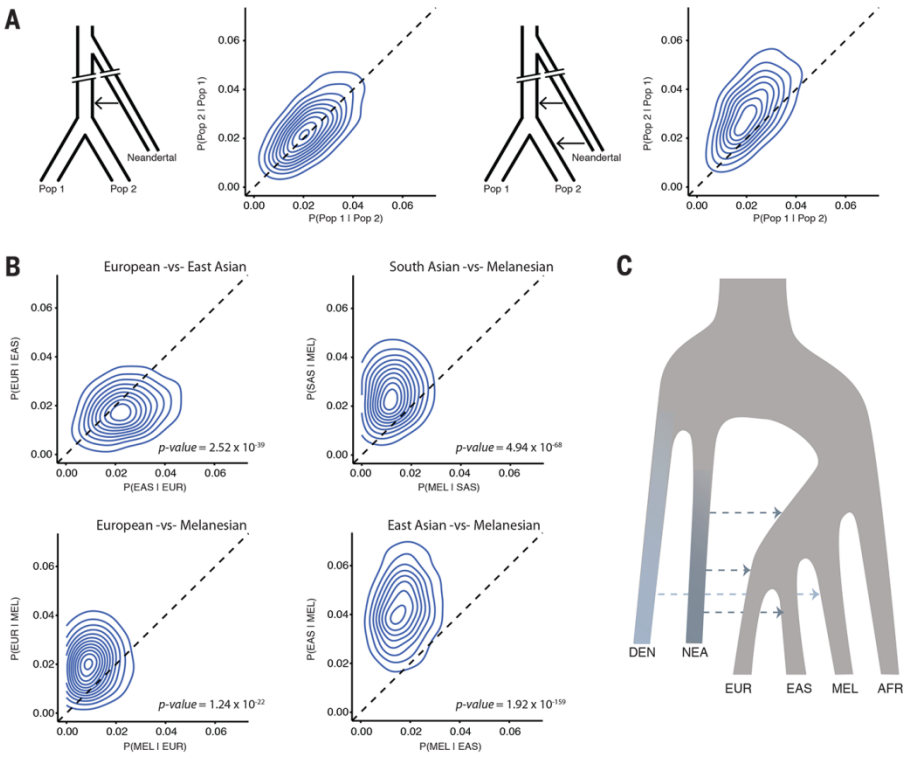


Fig. 3. Identifying shared and unique pulses of Neandertal admixture among human populations. (**A**) Schematics of two simulated introgression models, and patterns of reciprocal match probabilities. Contour plots are fit to the scatterplot of reciprocal match probabilities calculated from analyzing all pairwise combinations of individuals between two populations. Left, gene flow occurs into the common ancestor of Population 1 and Population 2, and reciprocal match probabilities fall along the diagonal as predicted by theory (Binomial test, $P > 0.05$) (*14*). Right, Population 2 receives additional admixture shifting reciprocal match probabilities above the diagonal (Binomial test, $P < 0.05$). (**B**) Reciprocal match probabilities of Neandertal sequences in modern human populations, consistent with additional Neandertal admixture into East Asians versus Europeans, and into Europeans, East Asians and South Asians versus Melanesians. (**C**) Schematic of admixture history consistent with the data.

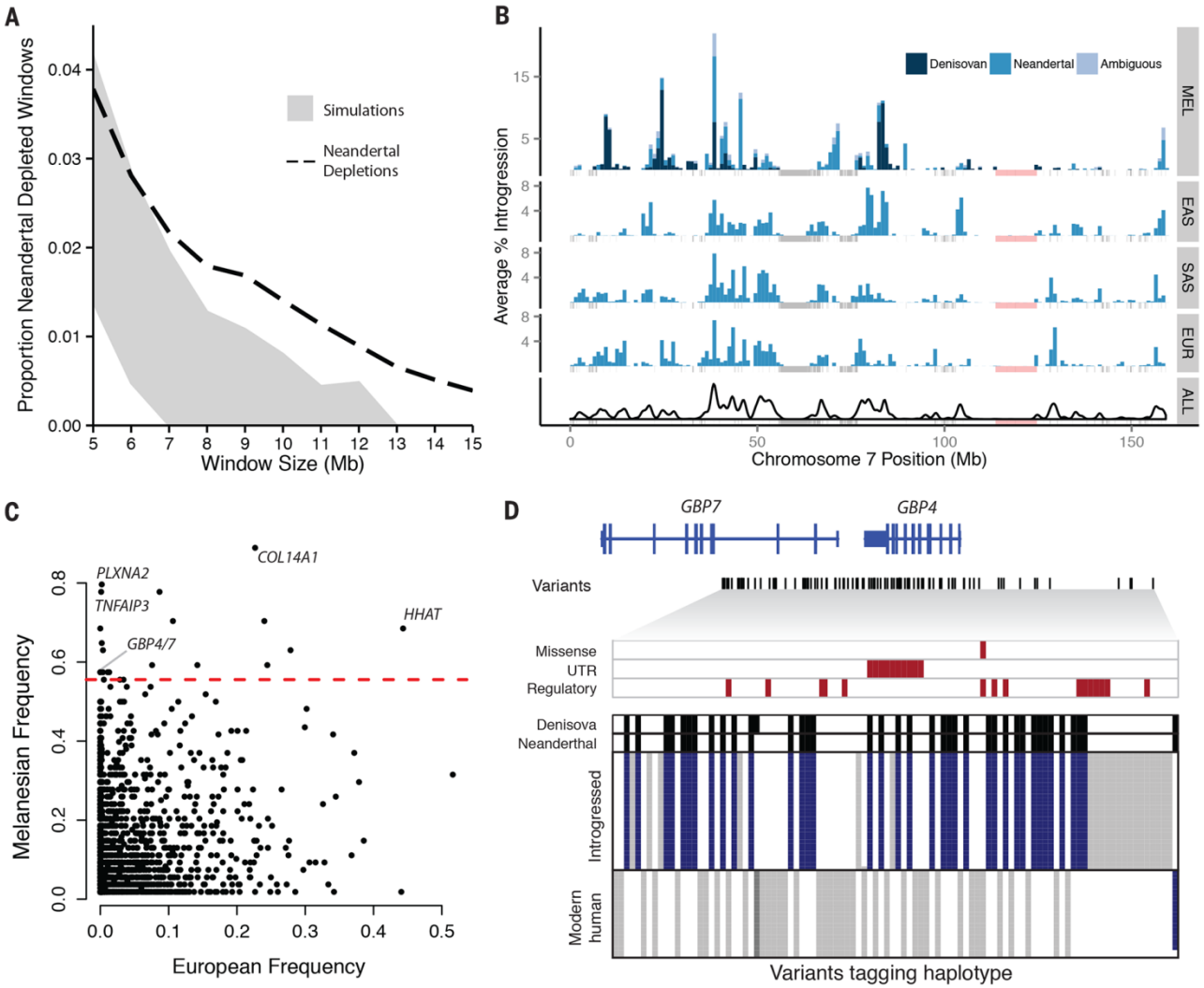


Fig. 4. Maps of archaic admixture reveal signatures of purifying and positive selection. (**A**) Proportion of windows significantly depleted of Neandertal introgression in Europeans and East Asians (dashed line) versus what is expected in neutral demographic models (95% CI in gray). (**B**) Distribution of Neandertal and Denisovan sequence across chromosome 7 in Melanesians (MEL), East Asians (EAS), South Asians (SAS), Europeans (EUR), and summed across all populations (TOT). Masked regions are shown as grey vertical lines. An 11.1 Mb region significantly depleted of Denisovan and Neandertal ancestry in all populations is shown in light gray. (**C**) The frequency of archaic haplotypes in Melanesians versus Europeans. The red line indicates the 99th percentile defined by neutral coalescent simulations. Notable genes are labeled. (**D**) Visual representation of a high frequency haplotype encompassing *GBP4* and *GBP7*. Rows indicate individual haplotypes and columns denote variants that tag the introgressed haplotype (*14*). Alleles are colored according to whether they are ancestral (white), derived variants that match both archaic genomes (blue), derived variants that match one archaic genome (dark grey), or are derived but do not match either archaic genome (light gray). Archaic sequences are represented above, with black denoting derived variants. Missense, UTR, and putative regulatory variants (*14*), are highlighted with red boxes.